

Distribution plots:

* EDA concept we use mainly

* Essential in EDA Concepts exploratory data analysis

* Shows how data's are distributed

* show how the values are distributed in a dataset - data spread, most frequent values, skewness

→ Ess

⇒ histplot

⇒ kdeplot

⇒ displot

Histplot:

a histogram divides numerical data into intervals called bins and display the frequency of values in each interval.

Syntax:

sns.histplot(data=df, x="column")

ex:

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
df = sns.load_dataset("tips")
```

```
sns.histplot(data=df, x="total_bill")
```

```
plt.show()
```

Tall ⇒ more customer bill range

bins parameter pass

```
data=df, x="total_bill", bins=20
```

Smaller bin → more detail

change color

```
color="green"
```

Kde plot:

* Kernel density estimation plot is a smooth curve that estimate the probability density of the data. Unlike a histogram, it has no bars.

Syntax:

```
sns.kdeplot (data = df, x = "column")
```

ex:

```
sns.kdeplot (data = df, x = "total_bill",  
             fill = True)
```

Interview qn:

Histogram

uses bars
Shows Frequency
easier to Count values

Kde plot

uses a smooth curve
Shows probability density
easier to identify
overall distribution

Displot:

It is a figure level function, used to visualize the distribution of data. It can create histogram, kde plot on both

ex:

```
import seaborn as sns  
import matplotlib.pyplot as plt  
df = sns.load_dataset("tips")  
sns.displot (data = df, x = "total_bill")
```

```
plt.show()
```

~~bins = 15~~
use bins.

Using kde & displot:

```
sns.displot (data = df, x = "total_bill",
```

```
kde = True)
```

```
plt.show()
```


Outliers:

- Isolated area (or) isolated points
- ex: park $\Rightarrow$ Groups of people, one man separate
 $\Rightarrow$ that man is isolated
 $\Rightarrow$ 3rd party is isolated.

* is a data point that is significantly different from the rest of data; an outlier is a value that is far away from most of the data.

qns:

1. Load any dataset
2. Create a histogram for a column
3. Create a histogram with bins
4. Create a filled kdeplot
5. Create a displot.]

Box plot:

Displays the distribution of numerical data and helps identify

Median, quartiles, Spread of data, outliers.

ex:

```
import seaborn as sns
import matplotlib.pyplot as plt
df = sns.load_dataset("tips")
sns.boxplot(data=df, x="day", y="total_bill")
plt.show()
```

violin plot:

Combines a boxplot with a kde plot
median, distribution, density, spread.

ex:

```
sns.violinplot(data=df, x="day", y="total_bill")
```

Strip plot:

Displays every individual data point.

Similar to scatter plot.

ex: `sns.stripplot(data=df, x="day", y="total_bill")`

Jitter:

Slightly spread overlapping points so that all observation became visible.

ex: `sns.stripplot(data=df, x="day", y="total_bill", jitter=True)`

Swarm plot:

It similar to strip plot but automatically arranges the points so they do not overlap. Unlike strip plot points are adjusted automatically to avoid overlapping.

ex: `sns.swarmplot(data=df, x="day", y="total_bill")`

Box plot Use here:

`sns.boxplot(data=df, x="day", y="total_bill", hue="sex")`

detect outliers → box plot

view density distributions → violin plot

Display every observation → Swarm plot without overlap

In:

load any dataset?

Create a boxplot with 2 columns (vs)

Create a violin plot

Create a strip plot with jitter

iris dataset:

columns:

sepal - length
sepal - width
petal - length
petal - width
Species.

Tips:

total_bill
tip
Sex
Smoker
day
time
size

Penguins

species
bill - length - mm,
bill - depth - mm
flipper - length - mm
body mass - g
Sex

diamonds

carat
cut
color
clarity
price

Flights

Year
month
passenger

Titanic:

Survived
Sex
age
class
fare
embarked

~~tips~~